

An Anchored Field-Target Update for Critic-Guided Fine-Tuning of Generative Policies

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Abstract

Behaviour-cloned robot policies are capped by their demonstrations, and the usual way past that cap is to train a critic and fine-tune against it. This reliably goes wrong: the policy drifts away from the data the critic was fit to, its score rises, and its actual success collapses.

We present an actor update that does not. A frozen base proposes a cloud of candidate actions; each is displaced along the critic’s gradient; a restoring force returns any candidate that has strayed beyond a radius ρ of the base; and a small residual head is regressed onto the displaced candidates. Nothing else in the pipeline changes — the critic, replay, demonstrations, budget and evaluation are those of the state-of-the-art residual fine-tuner, byte for byte, so the only variable is the actor update.

Under that control, the update gains +9.4 points on the eight hardest LIBERO-90 tasks where backpropagating $-Q$ gains +0.9, and lifts robomimic square from 0.382 to 0.91–0.93. It reaches the published flow-matching pipeline’s performance from a base less than half as strong, and transfers unchanged to that flow-matching base.

We then ask which part is responsible, switching each guard off independently on two tasks. The answer is decisive in one direction only. Removing both is catastrophic: success falls to exactly 0.000 on both tasks, from bases of 0.573 and 0.610, so the unconstrained action-gradient update — what prior work in this family runs — does not merely underperform here, it destroys the policy. But either guard alone recovers essentially all of the gain (0.671–0.803 across configurations, against 0.781 and 0.684 for both together), and the gaps between them are smaller than this benchmark’s seed-to-seed spread. We therefore claim only that a constraint tying the policy to its base is necessary; which constraint, at this budget, we cannot distinguish.

1 Introduction

A behavior-cloned robot policy is only as good as its demonstrations. To push past them, the usual move is to learn a critic and fine-tune against it with reinforcement learning. This paper is about one design choice inside that loop, which we think is made wrongly almost everywhere: how much authority the critic is given.

A critic does two jobs in a standard update. It says which way the policy should move, and — through the magnitude of its gradient — how far. Only the first is a job it was trained for. A critic learns to rank actions in the region its data covers; outside that region nothing constrains it, and it is quite capable of being steep and wrong at the same time (Fujimoto et al., 2018; 2019; Jain et al., 2025). The consequences of the two errors are not comparable. A slightly wrong direction gives a slightly wrong step. A wrongly large distance takes the policy off the data manifold in one update, and the further off it goes the better it looks to the critic that sent it there.

Our setting sharpens the problem, because it removes the usual escape route. We fine-tune one-step generative policies — a drifting model (Deng et al., 2026) maps noise to a

multimodal action in a single network evaluation — and such a policy has no tractable likelihood, so the ratio-based toolbox is unavailable. It can be recovered only by injecting noise into the deployed policy (Zhang et al., 2026), which changes what ships, or by assuming a denoising objective the policy does not have (McAllister et al., 2025) (Section 3.2). With likelihoods off the table, a critic is what remains — which makes how we consume it the whole question.

Both standard answers delegate the distance. Backpropagating $-Q$ through the actor, as the state-of-the-art residual fine-tuner DICE-RL (Sun & Song, 2026) does, makes every parameter step proportional to $\|\nabla_a Q\|$. The action-gradient family (Li et al., 2024; Psenka et al., 2025) never differentiates through the policy, but moves each sampled action to wherever a gradient step lands and regresses onto that — the same delegation, one level up. A controlled toy (Section 3.3) shows how bad this can get: with high-dimensional actions and scarce critic data, the backpropagated update drives true reward to zero while its own Q -estimate rises by four orders of magnitude. A behavior-cloning penalty does not save it, because a penalty in the loss must out-shout an unbounded gradient, and loses. Yet the gradient is not the problem. In the opposite regime, where the critic generalizes, it is the best signal available, and methods that discard it waste it. What is needed is a rule that keeps the direction in both regimes while capping how far any single step may carry the policy.

The lesson we draw is narrow and we state it as such: a critic may be trusted with the direction of a policy update, and with its length only up to a bound. We are not claiming the critic should be stripped of the step size entirely — in our own method it still sets it for the majority of sampled actions (Section 5.4), and that is fine. The claim is that the tail must be truncated, because that is where the critic is least constrained and where a single step can end the run. Discarding the direction, as critic-as-judge methods do (Peng et al., 2019; Wang et al., 2020), throws away the best signal available when the critic is trustworthy; letting the critic set the length hands it a quantity it was never trained to supply. Note that this indicts more than backpropagation. The action-gradient family — which improves actions along $\nabla_a Q$ and regresses the policy onto the improved actions (Li et al., 2024; Psenka et al., 2025) — keeps the direction but still leaves the distance to the critic, and is reported to degrade exactly as one would predict, drifting off the behavior distribution when the critic’s gradient is biased (Choi et al., 2026). What has been missing across both is not a better signal but a bound, and one placed where the harm occurs: in action space, not parameter space. Gradient clipping constrains the parameter step, which says nothing about how far the resulting action moves.

Porygon supplies that bound. A frozen base proposes action particles through a residual head, and reward enters as a velocity field V_Q — by default the normalized critic gradient. Two guards then act before anything is regressed. First, each particle’s total displacement is clipped pointwise in action space, bounding how far the regression target may move — though not, as we are careful to establish in Section 4.1, how far the network’s output actually moves once those targets are amortized through shared parameters. Second, a restoring anchor field with a dead zone pulls the residual back toward the base whenever it strays beyond a radius ρ , leaving it untouched inside. Only then is the residual regressed onto the displaced targets, and because the update rewrites nothing in the replay buffer, stored transitions remain consistent with the dynamics that produced them. No likelihood is ever needed, and in the small-step limit the update recovers the backpropagated direction exactly (Proposition 1) — so the safety costs nothing asymptotically.

The construction is also economical on the policy class that motivated it. A drifting model (Deng et al., 2026) is pretrained by field-target regression — transport samples along a velocity field, regress the network onto the transported targets — so on that base the fine-tuner reuses machinery the policy already has. That economy is a convenience, not a requirement: the update assumes only a frozen noise-to-action map, and we verify it transfers unchanged to a multi-step flow-matching base.

Because reward enters only as a field, the critic can also be consumed at lower resolution when its gradients cannot be trusted: transport toward critic-ranked top- k particles, or toward the cloud’s own $\exp(Q/\tau)$ -tilted density. Our experiments treat this as an explicit

dial and map its regimes. On robomimic, where the critic generalizes, the gradient field is the resolution to use: from a drifting base at 0.877, Porygon reaches 0.99 on can, and lifts square from 0.382 to 0.91–0.93 across seeds — ahead of the backpropagated control at the matched budget (0.87–0.91) and at parity with the flow-matching + DICE-RL state-of-the-art pipeline on both tasks, while the value-only sources trail far behind (~ 0.55 on square). On the eight hardest LIBERO-90 tasks, fine-tuned per task from one shared base with one shared recipe, Porygon gains +3.9 to +23.9 points (mean +9.4) where the backpropagated actor consuming the same critics gains +0.9 and loses ground on the two highest-headroom tasks. The ordering is unchanged when the base is swapped for the stronger flow-matching policy, and Porygon matches a critic-free policy-gradient baseline at equal budget without needing the noise injection that baseline requires. With a single critic shared across all eight tasks, no critic-based update rule — backpropagated or field-based, ours or the state of the art’s — improves on its own base, locating the binding constraint in per-task critic quality exactly as the toy predicts.

Our contributions:

- An action-space trust region for action-gradient policy improvement (Section 4). Improving actions along $\nabla_a Q$ and regressing the policy onto them is an established family (Li et al., 2024; Psenka et al., 2025; Choi et al., 2026), and we adopt that form unchanged. What the family lacks is anything holding the policy near the data, and the obvious candidate — a behaviour-cloning penalty applied at every step — is the wrong shape, because it opposes the critic in the region where the critic is reliable; measured, it lands at base level (0.549 and 0.601 against bases of 0.573 and 0.610). We therefore place the constraint where the harm occurs, in action space, and we test two forms of it: a pointwise clip on each particle’s displacement, and a restoring anchor that is silent within a radius ρ of the frozen base and engages only beyond it. We are deliberate about what this buys. Removing both is catastrophic — success falls to exactly 0.000 on both tasks — while either one alone recovers essentially all of the gain, and deleting the anchor entirely is not merely survivable but among our best settings (0.803 on t65). We therefore claim that some action-space constraint is necessary, and explicitly not that ours is the right one (Section 5.4). Two further choices separate our instantiation from the family: targets are formed from freshly drawn particles off a frozen base rather than from buffer actions mutated in place, so stored transitions stay consistent with the dynamics that produced them, and the update needs no likelihood.
- A controlled diagnosis of critic exploitation (Section 3.3): a two-regime toy showing the failure is real, regime-dependent, and not fixed by BC penalties — motivating trust regions in action space rather than loss space.
- Evidence that the form, not the signal or the step size, is what wins (Section 5): holding base, critic, data and budget fixed and varying only the actor update, Porygon beats the backpropagated actor on both benchmarks and on both bases, lifts a weaker one-step base to match the stronger flow-matching + DICE-RL pipeline, and reaches parity with a critic-free policy-gradient baseline without modifying the deployed policy. Two controls close the obvious alternative explanations. Sweeping each method’s own step size an order of magnitude leaves the two in non-overlapping bands — the backpropagated actor never rises above its base at any learning rate, ours never falls near it at any reward step — so the gap is not tuning. And deleting only the critic’s gradient direction, keeping everything else, collapses performance below the untuned base on two separate tasks — so the gain is not simply caution.

2 Related Work

RL fine-tuning of generative robot policies. Policy-gradient fine-tuning of vision-language-action models with sparse task reward is now standard for autoregressive token policies (Li et al., 2025), but transfers poorly to continuous generative heads: likelihood ratios are intractable for diffusion and flow policies, and workarounds either reformulate the denoising chain as an MDP (Ren et al., 2025; Black et al., 2024) — a construction with no analog for

one-step generators — or avoid likelihoods entirely (Park et al., 2025; Zhang et al., 2026; Mark et al., 2024). FPO (McAllister et al., 2025) is the strongest form of the likelihood-free ratio: it replaces $\pi_\theta/\pi_{\theta_{\text{old}}}$ with the exponentiated difference of per-sample conditional flow-matching losses, which the CFM-ELBO relationship makes a valid proxy, and it does so without perturbing the deployed policy. It is therefore the right policy-gradient method for a flow or diffusion policy — but the surrogate is defined by a denoising loss over a flow time τ , and a one-step drifting model has neither: its training objective is field-target regression, with no ELBO to bound. FPO also fine-tunes (or trains from scratch) the full policy on continuous-control suites, where we update a residual on a frozen base for manipulation. The two are complementary — FPO makes the ratio tractable, we make the critic safe to follow — and we regard evaluating FPO on a flow-matching base against our update as the clearest missing comparison in this paper. Closest to our setting, DICE-RL (Sun & Song, 2026) fine-tunes a frozen one-step distillation of a flow-matching policy by training a residual head with a backpropagated critic and a filtered BC anchor; it is our primary baseline and the control for our actor-update ablation, since we keep its critic, residual parameterization, and data pipeline unchanged.

Where this update sits. RL for diffusion and flow policies divides into four families (Choi et al., 2026): backpropagating through the denoising chain (Diffusion-QL, and DICE-RL’s residual actor); ratio-based methods that make a likelihood tractable (DPPO, FPO); Q -weighted regression; and action-gradient methods, which improve actions with $\nabla_a Q$ and then regress the policy onto the improved actions. Porygon belongs to the last family, and we want to be exact about what is and is not new in it. The primitive of displacing actions along $\nabla_a Q$ and regressing onto the result is not ours: DIPO (Li et al., 2024) introduced it, updating replay-buffer actions by gradient ascent and fitting a diffusion policy to them, and QSM (Psenka et al., 2025) reaches the same signal differently, by matching the policy’s score to $\nabla_a Q$. What the family lacks — and what the survey notes none of its members supply — is any bound on how far that displacement may carry the policy, or any restoring force toward a behavior policy. That gap is also its documented weakness: action-gradient methods are reported to degrade when the critic’s gradient is biased, and to drift away from the behavior distribution.

Our contribution is precisely that missing piece, and our ablations measure it directly. The displacement is clipped pointwise in action space, a dead-zone anchor restores toward a frozen base, and the update is applied to a residual rather than by overwriting stored transitions (DIPO’s buffer rewrite makes the stored action inconsistent with the transition that produced it). Removing either guard reproduces the family’s failure inside our own system: performance falls below the untuned base (Table 5). Test-time guidance (Zhou et al., 2026) applies the same $\nabla_a Q$ signal without training at all, and is complementary.

Critic-as-compass versus critic-as-judge. That a learned Q -function’s action-gradient cannot be trusted is a recurring diagnosis: overestimation and narrow spurious peaks in TD3 (Fujimoto et al., 2018), extrapolation error in BCQ (Fujimoto et al., 2019), and spurious local optima that make sample-and-select beat gradient ascent (Jain et al., 2025). One response treats the critic only as a judge: weighted regression and sample-based improvement (AWR, CRR, MPO, IDQL, QT-Opt) (Peng et al., 2019; Wang et al., 2020; Abdolmaleki et al., 2018; Hansen-Estruch et al., 2023; Kalashnikov et al., 2018) tilt the policy toward $\pi \cdot \exp(Q/\tau)$ using only Q -values. Our method spans this spectrum rather than picking a side: the update form is judge-safe (bounded transport toward targets), while the reward field inside it can still be the analytic gradient where it is trustworthy — and our ablations map exactly when each choice wins.

Particle transport as a policy update. Regressing a sampler onto displaced particles is the amortized-SVGd idea of Soft Q -Learning (Haarnoja et al., 2017), with kernel repulsion preserving diversity as in SVPG (Liu et al., 2017); drifting models (Deng et al., 2026) make the same operator the pretraining objective of a one-step generator. Drifting Field Policy (Koo et al., 2026) is the closest prior method: it also performs RL on a drifting backbone by drifting fresh policy samples toward critic-ranked top- K candidates. Porygon differs in three ways: we update a residual on a frozen base rather than the full policy, we transport

with an explicit pointwise trust region and a restoring anchor field (the stability contract our toy shows is load-bearing), and our reward field is a dial — including the analytic $\nabla_a Q$ source that dominates when the critic generalizes — rather than a fixed ranking surrogate.

Residual RL and staying on-manifold. Learning a residual on a frozen base is a long-standing recipe for sample-efficient robot RL; DAWN (Ma et al., 2026) identifies the value cold-start and scale pathologies of residual critics that we also observe. The off-manifold drift our toy exhibits is the mechanism behind over-optimization and mode collapse in RL-finetuned generators (Black et al., 2024; Fan et al., 2023; Nakamoto et al., 2023), and forgetting in RL-finetuned language models is likewise governed by divergence from the base (Shenfeld et al., 2025) — which cannot even be measured without likelihoods, reinforcing update rules whose step size is bounded in action space by construction.

In sum, prior work either trusts the critic’s gradient and pays for it off-manifold, or discards it and pays on hard tasks; Porygon is, to our knowledge, the first fine-tuner for one-step generative policies whose update form is safe under an arbitrary critic while still admitting the full first-order signal when it is available.

3 Problem Statement

3.1 Setup

We consider an episodic decision process with states s , action chunks $a \in \mathbb{R}^{H \times d}$ (H control steps of a d -dimensional action, treated as a single vector of dimension $S = Hd$), and sparse task-success reward. A one-step generative policy maps a state and a noise sample to an action in a single network evaluation,

$$a = \pi_0(s, z), \quad z \sim \mathcal{N}(0, I), \quad (1)$$

and is pretrained on demonstrations with a drifting-model objective (Deng et al., 2026): at each step, a population of policy samples is transported by a kernel velocity field toward the data distribution, and the network is regressed onto the transported targets. We write $\mathcal{V}(x \mid p^+, p^-)$ for this transport field acting on query particles x , attracted to particles of p^+ and repelled from particles of p^- (Section 4.2 recalls its exact form).

Following residual fine-tuning practice (Ma et al., 2026; Sun & Song, 2026), we freeze the base and learn a residual head on top of it,

$$a = \pi_0(s, z) + r_\theta(s, z), \quad (2)$$

together with an ensemble critic $Q_\phi(s, z, a)$ trained by temporal difference with a conservative (CQL-style) penalty (Kumar et al., 2020). The critic training is standard and identical for every method compared in this paper; the object of study is solely the actor update — how the policy consumes the critic.

3.2 Two structural failure modes

No likelihoods. Policy-gradient fine-tuners (PPO/GRPO-style, e.g. SimpleVLA-RL (Li et al., 2025)) require $\log \pi(a \mid s)$. A one-step generator defines its action distribution implicitly through the noise map Equation (2); its likelihood is intractable, and unlike multi-step diffusion policies there is no chain-MDP reformulation to fall back on (Ren et al., 2025; Park et al., 2025). Two routes recover a ratio, and neither is available to a one-step drifting policy. The first modifies the policy — adding fixed-variance Gaussian noise to its output so that a surrogate log-density exists (Zhang et al., 2026) — so the deployed policy is no longer the deterministic one-step map that was pretrained; we include this surrogate as a baseline and it is a competitive one. The second leaves the policy untouched but requires it to be a flow model: FPO (McAllister et al., 2025) forms a PPO-clip ratio from differences of per-sample conditional flow-matching losses, exploiting the CFM–ELBO relationship. That construction needs a denoising loss defined over a flow time τ ; a drifting model has no τ and is trained by field-target regression, which carries no such ELBO link, so the surrogate does not exist for it. On a flow-matching base FPO is applicable and would be the natural

policy-gradient comparison — a comparison we do not run here. The update we develop needs no likelihood, surrogate or exact, on either base.

Letting the critic set the distance. The remaining updates all use the critic’s action-gradient, and all let its magnitude decide how far the policy moves. Backpropagating $-Q$ through the actor, as in the DDPG lineage and in DICE-RL’s residual actor (Sun & Song, 2026), does so explicitly:

$$\nabla_{\theta}\mathbb{E}[-Q] = -\nabla_a Q \cdot \partial a / \partial \theta, \quad (3)$$

so a steeper critic produces a larger parameter step. The action-gradient family — which improves each sampled action by a gradient step and regresses the policy onto the improved action (Li et al., 2024; Psenka et al., 2025) — avoids differentiating through the policy, but inherits the same coupling one level up: the improved action sits wherever $a + \alpha \nabla_a Q$ lands, so the displacement is again proportional to the critic’s steepness. Reported failures of that family match the prediction, degrading under biased critic gradients and drifting off the behavior distribution (Choi et al., 2026).

Why is distance the risky thing to delegate? A critic is trained to rank actions near its data. Away from that data it is unconstrained, and can be steep and wrong at once (Fujimoto et al., 2018; 2019; Jain et al., 2025). The two quantities then fail asymmetrically. A slightly wrong direction costs a slightly wrong step. A wrongly large distance costs the data manifold: the policy is carried off it in one update, and the better it then scores under the critic, the worse it actually is.

The important word is persistent. The individual steps need not be large; what matters is that they keep pointing the same way. The critic’s extrapolated optimum does not move as the policy approaches it, so each update advances toward the same place and nothing returns the policy. This distinction turns out to decide which remedy works: limiting the size of a step does not stop a drift assembled from permitted steps, whereas a force that returns the policy does (Section 5.4).

3.3 A diagnostic experiment

To see the failure cleanly we build the smallest setting that contains it: a bandit whose expert data has two modes, a one-step policy pretrained on that data with the drift operator, and a critic trained only on actions the expert actually takes. The critic is therefore accurate on the data and unconstrained everywhere else — the situation every offline-trained critic is in. We then fine-tune the same base with two update rules and vary one thing: the action dimension, which controls how much of the space the critic has never seen.

When the critic is reliable, follow it. With 2-D actions and dense critic data, backpropagation is the best update (0.79 true reward, versus 0.53 for a value-only field update and 0.50 for the base). Nothing is wrong with using a critic’s gradient when the critic knows what it is talking about.

When it is not, backpropagation is catastrophic — and a BC penalty does not save it. With 16-D actions and 60 critic training points, the same update drives true reward to zero while the critic’s own estimate of that policy rises by four orders of magnitude: the policy has walked into a region the critic scores highly and has never seen. Adding the standard behavior-cloning penalty changes nothing (0.00 as well). This is the diagnostic result of the paper: the penalty is a term in a loss, and it is competing against a gradient of unbounded magnitude. It loses. Figure 1 shows the geometry — the policy is not wandering, it is being pulled, quickly and directly, toward the critic’s hallucinated optimum.

Which fix actually works. The natural cure is to limit how far the policy may move per update, and we tried it. It does not work: with the step bound alone the policy still ends at zero true reward, $5,847\times$ the data’s own spread away — no better than applying no guard at all. The reason is visible in Figure 1(a). The critic’s imagined optimum does not recede as the policy approaches it, so every permitted step points the same way, and a drift assembled from small permitted steps arrives just as surely as one large one. A speed limit is not a boundary.

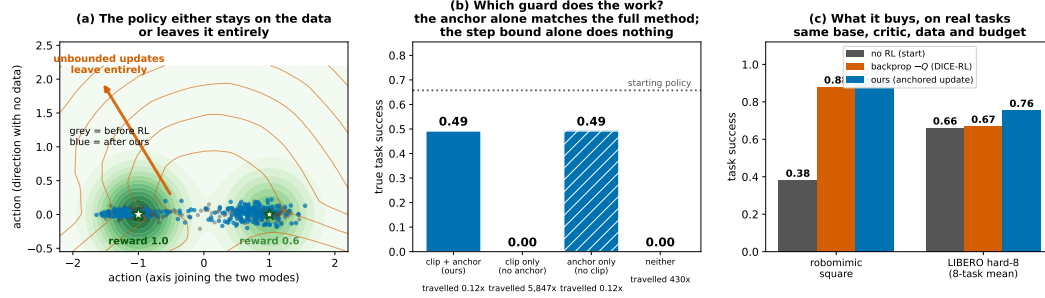


Figure 1: Which guard keeps a critic-driven update on the data? Panels (a)–(b) use a diagnostic bandit: 16-D actions, two demonstration modes of reward 1.0 and 0.6, and a critic deliberately fit to only 60 expert actions so that it is wrong away from them. We use it because the entire action space can be drawn, not because it resembles a robot task. (a) A slice of that space. Green is the true reward, thin orange lines are what the critic believes, and the two part company away from the data. Our update (blue) leaves the policy’s actions on the demonstrated modes; unbounded updates exit the frame entirely. (b) The two guards, switched on and off independently. A step bound on its own does nothing — 0.00 success, and $5,847\times$ the data’s spread away, no better than using no guard at all — because a drift assembled from many permitted small steps is still a drift. The dead-zone restoring force on its own reproduces the full method exactly (0.49, $0.12\times$). This identifies the mechanism: what matters is the pull back toward the starting policy, not the limit on step size. (c) What that buys on real benchmarks, with base, critic, data and budget held fixed: robomimic square rises from 0.38 to 0.92, and the eight hardest LIBERO-90 tasks gain +9.4 points against +0.9 for the backpropagated actor. Note that in (b) no configuration exceeds the untuned base (0.66): the diagnostic is built to separate the guards, not to show improvement, which is the job of (c).

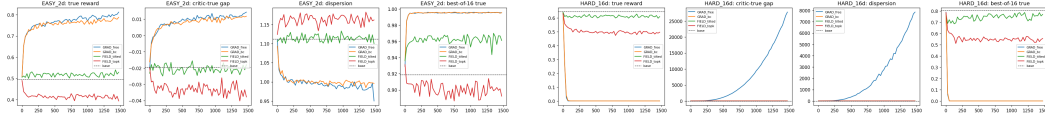


Figure 2: The two-regime toy (3 seeds). Left, easy regime (2-D actions, dense critic data): the critic’s gradient generalizes and backpropagation ($-Q$, with or without a BC anchor) is the best update. Right, hard regime (16-D actions, 60 critic points): the same updates collapse to zero true reward while their internal Q -estimate explodes — the BC anchor does not help — whereas the bounded field-target update stays on-manifold and holds the base policy. A useful fine-tuner must keep the gradient where it is trustworthy and survive where it is not.

What works is a restoring force that is silent within a radius ρ of the base and engages beyond it. On its own, with no step bound at all, it reproduces the full method exactly (0.49 true reward, $0.12\times$ the data spread), and adding the step bound on top changes nothing (Figure 1b).

Three conclusions carry into the method. First, the critic’s gradient is the best signal available where it generalizes and a liability where it does not, so a useful update keeps the direction. Second, what protects the policy is a force that returns it, not one that slows it: containment is a statement about where the policy ends up, not how fast it travels. Third — and this is why the dead zone rather than a plain penalty — the restoring force must be absent near the base, or it spends its strength opposing the critic in the one region where the critic is reliable. A constant behaviour-cloning penalty does exactly that, and in the same setting it also ends at 0.00.

[Figure reserved: method overview.] Left: the frozen base π_0 maps noise particles z_i to a cloud of actions; the residual r_θ shifts them. Middle: the three velocity fields at each particle — clipped critic gradient V_Q (with the top- k / tilted alternatives inset), dead-zone anchor V_{anc} — and the total-motion clip δ_{tot} . Right: the regression of r_θ onto the displaced targets; the same schematic as drifting pretraining with one extra field. To be drawn (TikZ or Figma).

Figure 3: Porygon overview: reinforcement fine-tuning as one more velocity field in the pretraining transport. [TODO: draw]

4 Method

Section 3.3 left us with a concrete requirement: follow the critic where it has data, and prevent the policy from wandering to where it does not. Crucially, this is a requirement about where the policy ends up, not about how fast it travels — a distinction the experiments turn out to hinge on.

The vehicle for that bound is an existing primitive, which we adopt rather than claim: move action samples along a velocity field, then regress the network onto the moved samples. This is how the action-gradient family improves a policy (Li et al., 2024; Psenka et al., 2025), and — not by coincidence — also how a drifting model is pretrained (Deng et al., 2026), so on such a base fine-tuning is one more field added to a transport the policy class already performs. What that family does not do, and what this section contributes, is constrain the displacement: in DIPO the improved action is wherever the gradient step lands, and the policy is then regressed onto it.

The contribution is the anchor field V_{anc} : a pull back toward the frozen base that is silent while the residual stays within a radius ρ and engages only outside it (Section 4.3 gives its form). It is not an ad hoc construction — it is the negative gradient of a hinged radial potential, and at $\rho = 0$ it reduces to the ordinary quadratic residual regularizer the back-propagated control already uses. The dead zone is thus a hinge generalization of standard L_2 residual regularization, and the experiments say that where ρ sits on that interpolation decides whether fine-tuning works at all (Section 5.4).

The reward field V_Q says which direction improves each action. It is norm-clipped, and we keep the clip, but we do not attribute the method’s behaviour to it: switching the clip off while retaining the anchor leaves performance unchanged, whereas switching the anchor off while retaining the clip removes the gains entirely (Section 5.4).

Neither field is a loss term; both are displacements applied to actions before the regression. This does not make the anchor invincible — where $\|V_Q\| > \|V_{\text{anc}}\|$ the net displacement still points outward — but it means the two compete as bounded magnitudes in action space rather than through an unbounded parameter gradient, as a behaviour-cloning penalty must (Section 3.3). No likelihood is required, and no critic gradient is ever backpropagated into the actor.

4.1 The field-target update

One update works as follows. For each state in the batch we draw K noise samples and form the current action particles $a_i = \pi_0(s, z_i) + r_\theta(s, z_i)$, treated as constants (detached). Each particle is displaced by a combined velocity,

$$\tilde{r}_i = \text{sg}\left[r_i + \text{clip}_{\delta_Q}(V_Q(s, z_i, a_i)) + V_{\text{anc}}(r_i)\right], \quad \mathcal{L}_{\text{actor}} = \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2, \quad (4)$$

where $\text{sg}[\cdot]$ is stop-gradient and $\text{clip}_\delta(v) = v \cdot \min(1, \delta/\|v\|)$ rescales a field vector to norm at most δ .

In words, one update does four things. (1) Propose: draw K noise samples and push them through the frozen base plus the current residual to get a cloud of candidate actions for this state. (2) Score and displace: ask the critic which direction improves each candidate, clip that direction to length δ_Q , and add the anchor’s pull if the residual has strayed past its radius; clip the resulting total displacement to δ_{tot} . (3) Freeze the targets: the displaced actions are detached — they are data now, not a computation graph. (4) Regress: train the residual head to output those displaced targets. The critic is trained separately (Section 4.4); it is never differentiated through the actor, only queried.

Why this primitive, for this policy class. Field-target regression is not merely a safe update — it is the operation the network was shaped by. A step-distilled generator fine-tuned under a mismatched objective drifts from the distribution its one-step map encodes (Jiang et al., 2026; Yang et al., 2024); by keeping fine-tuning inside the pretraining operator family, Porygon changes only where the targets sit, not what kind of function the network is asked to represent.

What the clip does and does not guarantee. The clip bounds target transport, and only that:

$$\|\tilde{r}_i - r_i\| \leq \delta_{\text{tot}} \quad \text{for every particle } i. \quad (5)$$

It is tempting to read this as a bound on how far the policy moves, and an earlier version of this work did. That inference is invalid. The targets are amortized through a shared network by a gradient step, so to first order the realized change at particle i is

$$\Delta r_i \approx \alpha \sum_j J_i J_j^\top (\tilde{r}_j - r_j), \quad J_i = \partial r_\theta(s, z_i) / \partial \theta, \quad (6)$$

and the Jacobian coupling across particles carries no guarantee that $\|\Delta r_i\| \leq \delta_{\text{tot}}$. This is precisely the objection we raise against parameter-space gradient clipping (Section 3.2), and it applies to target-space clipping too: bounding what the network is asked for does not bound what it does. A per-particle guarantee would require an output-space proximal step, exact interpolation, or a tabular actor — none of which we impose.

The distinction matters less than it might, because the clip turns out not to be the operative component in any case. A drift assembled from many individually permitted steps is still a drift, since the critic keeps pointing the same way and nothing returns the policy. Switching the clip off while keeping the anchor leaves the method unchanged; switching the anchor off while keeping the clip destroys it (Section 5.4). We therefore keep the clip for numerical safety and attribute the behaviour to the anchor.

4.2 The reward field V_Q

The default (and headline) reward field is the normalized critic gradient evaluated at each particle,

$$V_Q^{\text{grad}}(s, z, a) = \eta_Q \frac{\nabla_a \bar{Q}_\phi(s, z, a)}{q_{\text{scale}}}, \quad q_{\text{scale}} = \mathbb{E}[|\bar{Q}_\phi|], \quad (7)$$

where $\bar{Q}_\phi$ is the conservative ensemble aggregate. This is deliberately the same scalar the DICE-RL actor optimizes: in the small-step, unclipped limit, minimizing Equation (4) recovers the backpropagated update direction exactly (Proposition 1), so any performance difference is attributable to the update form — the anchored target regression — and not to a different reward signal.

The kernel transport operator. The lower-resolution reward fields below, the pretraining objective, and the distributional anchor variant all use the same drifting-model transport operator (Deng et al., 2026). Given query particles x , attractor particles p^+ (optionally weighted), and repulsor particles p^- , distances are normalized by a global scale and, for each kernel radius R in a small set, a soft mutual-matching affinity

$A = \sqrt{\text{softmax}_{\text{row}}(-d/R) \odot \text{softmax}_{\text{col}}(-d/R)}$ converts matched attractors into pulls and matched repulsors into pushes; forces are summed over radii to give $\mathcal{V}(x | p^+, p^-)$ (full form in Appendix C). Intuitively, $\mathcal{V}$ moves a particle cloud so its density flows toward p^+ and away from p^- — it needs only samples of the target, never a density or a gradient.

Because V_Q enters Equation (4) only as a velocity, the critic can be consumed at lower resolution when its gradients are not trustworthy, using only Q -values:

- zeroth-order (top- k): rank the K particles by Q and set $V_Q = \mathcal{V}(a | \text{top-}k \text{ particles, all particles})$ — the pretraining kernel transport toward the critic’s preferred candidates;
- tilted: let every particle attract with weight $K \cdot \text{softmax}_i(\text{adv}_i/\tau)$, transporting the cloud toward its own $\exp(Q/\tau)$ -tilted density, the canonical improvement target of Peng et al. (2019); Abdolmaleki et al. (2018).

These are not competing methods but settings of a dial — how much of the critic’s local structure the update trusts: full first-order information, ranking only, or soft value weights. Section 5 maps the regimes: the gradient source dominates wherever the critic generalizes, while the value-only sources close the gap precisely in the weak-critic regime predicted by Section 3.3.

4.3 The anchor field

The kernel reward fields vanish beyond their largest radius and the gradient field carries no notion of the data manifold, so Equation (4) alone would let the residual random-walk. The anchor is a restoring field with a dead zone,

$$V_{\text{anc}}(r) = -\eta_{\text{anc}} \left(1 - \frac{\rho}{\|r\|} \right)_+ r, \quad (8)$$

which is inert inside the trust radius ρ and pulls back only the excess residual norm beyond it. Equivalently, Equation (8) is the negative gradient of a hinged radial potential,

$$U(r) = \frac{1}{2} \eta_{\text{anc}} (\|r\| - \rho)_+^2, \quad V_{\text{anc}}(r) = -\nabla_r U(r), \quad (9)$$

so it removes exactly the radial excess and nothing else. Setting $\rho = 0$ gives $U(r) = \frac{1}{2} \eta_{\text{anc}} \|r\|^2$ — the quadratic residual regularizer of the backpropagated control — so the dead zone is a hinge generalization of that penalty, with ρ interpolating between them. Unlike a $\lambda \|r\|^2$ penalty — which competes with an unbounded gradient and loses (Section 3.3) — Equation (8) composes with the clipped reward field as a velocity, so the residual scale is set geometrically rather than by a loss-weight arms race. Heuristically, if the reward field were stationary, radially outward and saturated at $\|V_Q\| = \delta_Q$, balancing it against the anchor outside the dead zone gives $\delta_Q = \eta_{\text{anc}} (\|r\| - \rho)$, i.e. $\|r\| = \rho + \delta_Q/\eta_{\text{anc}}$. We use this only as a scale estimate when choosing η_{anc} : a learned critic’s field varies in direction and magnitude with r , so there may be several fixed points, cycles, or none, and we claim no unique equilibrium. pointwise pull $-\eta_{\text{anc}} r$, whose fixed point reproduces the DICE-RL anchor; the dead zone matters exactly when the reward field is weak (Section 5, LIBERO).

4.4 Relation to the backpropagated actor; critic

Proposition 1 (Small-step equivalence). At the current parameters, the gradient of Equation (4) is $\nabla_{\theta} \mathcal{L}_{\text{actor}} \propto -(\text{clip}_{\delta_Q}(V_Q) + V_{\text{anc}})^\top \partial r_{\theta} / \partial \theta$. With $V_Q = V_Q^{\text{grad}}$, no clipping ($\delta_Q, \delta_{\text{tot}} \rightarrow \infty$) and $\rho = 0$, this is the DICE-RL residual actor gradient with anchor weight η_{anc} .

Porygon is therefore a strict generalization of its control: the new degrees of freedom are the pointwise action-space clip, the dead-zone anchor, and the resolution dial on V_Q — each of which we ablate.

Remark (not just clipped gradient ascent). One might object that Equation (4) with V_Q^{grad} is “DDPG with clipping.” The distinction is where the bound lives. Clipping ∇_{θ} bounds a

Algorithm 1 Porygon: one update (V_Q^{grad} variant)

Require: frozen base π_0 ; residual r_θ ; critic Q_ϕ ; steps $\eta_Q, \eta_{\text{anc}}$; clips $\delta_Q, \delta_{\text{tot}}$; dead zone ρ ; particles K

- 1: sample batch of states s ; draw $z_i \sim \mathcal{N}(0, I)$, $a_i \leftarrow \pi_0(s, z_i) + r_\theta(s, z_i)$ $\triangleright$ no grad
- 2: $V_i \leftarrow \text{clip}_{\delta_Q}(\eta_Q \nabla_a \bar{Q}_\phi(s, z_i, a_i) / q_{\text{scale}})$ $\triangleright$ or top- k / tilted kernel field
- 3: $V_i \leftarrow V_i - \eta_{\text{anc}}(1 - \rho / \|r_i\|)_+ r_i$ $\triangleright$ dead-zone anchor, Equation (8)
- 4: $\tilde{r}_i \leftarrow \text{sg}[r_i + \text{clip}_{\delta_{\text{tot}}}(V_i)]$ $\triangleright$ policy moves $\leq \delta_{\text{tot}}$, always
- 5: $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2$
- 6: update ϕ : ensemble TD + CQL penalty $\triangleright$ unchanged from Sun & Song (2026)

parameter step, whose effect on the policy output is unknown and state-dependent; Equation (4) bounds the policy’s motion at every particle, which is the quantity that off-manifold failure depends on (Section 3.3). Moreover, only the regression form makes the two other components expressible at all: a dead-zone anchor and a value-only V_Q have no loss-gradient counterpart. The ablations in Section 5.4 test each distinction in isolation. The critic is unchanged from DICE-RL: ensemble TD with a CQL-style penalty pushing Q down at policy actions and up at data actions; field construction never leaks gradients into the critic (gradients w.r.t. actions only). Algorithm 1 summarizes one joint update.

5 Experiments

Our experiments answer four questions. Q1: Given the same critic, data and budget, does this actor update beat backpropagating $-Q$? Q2: Can a one-step drifting base fine-tuned this way match the stronger flow-matching pipeline? Q3: Which component is responsible — the restoring anchor, or the bound on step size? Q4: When should the reward field use the critic’s gradient rather than only its values?

5.1 Setup

Benchmarks. (i) robomimic can and square (state-based; 300-episode evaluations, average of the last three checkpoints; 3 seeds), run inside the official DICE-RL harness with only the actor update replaced — critic, replay, demonstrations, and evaluation protocol are byte-identical to the baseline. (ii) The eight hardest tasks of LIBERO-90 (multitask drifting base; 100 RL iterations; final numbers use powered evaluation: 100 rollouts $\times$ 3 evaluation seeds). [\[TODO: appendix table: env details, horizons, demo counts, budgets.\]](#)

Compared methods. Each comparison is chosen to isolate one of the three questions; every method on the drifting base shares the residual parameterization, critic, data, and budget, so only the actor update varies.

- Porygon: Equation (4) with the gradient reward field V_Q^{grad} .
- Backprop actor (the controlled comparison for Q1): the DICE-RL residual actor (Sun & Song, 2026) — backpropagated $-Q$ with a filtered BC penalty — run on our drifting base. It consumes the identical critic through the identical residual head, so any gap to Porygon is attributable to the update form, not to the base, the reward signal, or the data.
- FM + DICE-RL (the external reference for Q2): the unmodified state-of-the-art pipeline on its own flow-matching base (Sun & Song, 2026). It answers whether the one-step base plus Porygon competes with the strongest published system, not merely with its own control.
- Porygon (top- k) and Porygon (tilted) (resolution ablations for Q3, not baselines): the same update with the reward field built from the critic’s rankings (transport toward the top- k particles) or its soft values ($\exp(Q/\tau)$ -weighted self-attraction). Each deletes the critic’s gradient direction while changing nothing else, measuring

Table 1: robomimic success rate (300-episode evaluations, last-3-checkpoint average, per seed). Bold marks the best value in each column (ties within 0.005). “—” = matched-budget run still in progress at submission-draft time; no provisional numbers are shown. [†] these two baseline runs terminated before 20K steps, so their last-three-checkpoint averages are taken at 17–19K (s42) and 14–16K (s43); both curves are flat over that window, so the earlier window if anything favours the baseline.

method	can			square		
	s42	s43	s44	s42	s43	s44
drift base (no RL)		0.877			0.382	
backprop actor (same base)	0.970 [†]	0.946 [†]	0.940	0.870	0.909	0.861
FM + DICE-RL	0.989	0.979	0.986	0.934	0.923	0.930
Porygon (top- k)	0.924	0.920	0.910	0.562	0.525	0.562
Porygon (tilted)	0.949	0.947	0.919	0.578	0.584	0.560
Porygon	0.988	0.993	0.978	0.912	0.930	0.929

Table 2: Classic diffusion-RL fine-tuners on robomimic square, run from the official implementations without modification and from their own diffusion base (seed 42). These methods are not run at our budget: at their published settings each consumes $300 \times 400 \times 50 \times 4 \approx 24\text{M}$ environment steps, roughly $125 \times$ the 192K our own runs use, so the comparison is generous to them. Success is the same 300-episode protocol used throughout. “—” = training still in progress. Porygon is listed for reference only and starts from a different (drifting) base, which is why the base row for each column is given separately.

method	base it starts from	square (300 ep)
diffusion base (no RL)	—	0.279
DIPO (Li et al., 2024)	diffusion base	0.283
QSM (Psenka et al., 2025)	diffusion base	0.000
DQL	diffusion base	—
IDQL	diffusion base	—
AWR	diffusion base	—
Porygon (reference; drifting base 0.382)	drifting base	0.912–0.930

how much of Porygon’s gain flows through that one channel — and, in the weak-critic LIBERO regime, whether the gradient is worth consuming at all.

Hyperparameters are family-shared across Porygon and its two resolution variants, so comparisons isolate the update rule rather than tuning effort. One recipe is used per benchmark and never per task: $\delta_{\text{tot}}=0.15$, $\eta_{\text{anc}}=1.0$, $\tau=1$, $k=4$ throughout, with reward step $\eta_Q=0.5$ on robomimic and $\eta_Q=1.0$ on LIBERO, and a dead-zone radius $\rho=0.05$ in dimension-independent rms units on LIBERO (Section 5.4 shows why the units matter).

5.2 Robomimic: the gradient regime

Two of these have finished. DIPO ends at 0.283 against its own base of 0.279 — a gain of 0.4 points after 24M environment steps, inside evaluation noise. QSM collapses to exactly 0.000 in all six evaluation blocks. We checked the obvious explanation first: every hyperparameter in our QSM configuration matches the published one ($n_{\text{critic warmup}}=5$, actor/critic learning rates $10^{-5}/10^{-3}$, q_{grad} coefficient 10, $\tau=0.005$, buffer 5000, replay ratio 16, batch 1000), and we give it more training than published, not less (300 against 201 iterations). The collapse is also immediate rather than gradual: success is at base level (0.284) at iteration 0 and at zero by iteration 10, where it remains for the following 290 iterations. Both losses converge (actor 0.002, critic 0.01) rather than diverging, which is the signature of a degenerate fixed point — once the policy stops succeeding, every reward is zero, the critic correctly learns $Q \approx 0$ everywhere, and $\nabla_a Q$ carries no signal with which to recover. QSM aligns the policy’s score to $\nabla_a Q$ rather than adding a bounded correction to it, and nothing in the update ties the result to the base; this is the failure our constraint is meant to exclude. We note one

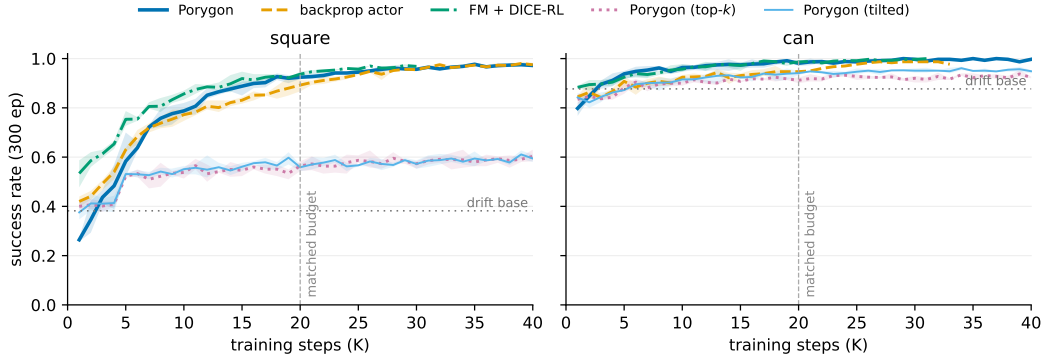


Figure 4: robomimic learning curves (300-episode evaluations every 1K steps; mean and min-max band over seeds; dashed vertical line marks the matched 20K budget). On square, Porygon overtakes the backprop actor by 7K steps and holds a 4–7-point lead through the matched budget; the value-only variants (top- k , tilted) flatline near 0.55. On can, Porygon and the flow-matching pipeline both approach the ceiling (≥ 0.98). [TODO: regenerate with the s44 and can-baseline reruns when they finish.]

caveat against reading too much into the magnitude: our diffusion base scores 0.279 where the original authors report roughly 0.4 on this task, and a weaker base implies a worse early critic, so we do not claim our 0.000 reproduces any published number.

Table 1 gives the per-seed results, and Figure 4 the corresponding learning curves. On can, fine-tuning saturates the task: Porygon reaches 0.978–0.993 from a base at 0.877, the flow-matching pipeline reaches 0.979–0.989, and the backprop actor trails slightly at 0.940–0.970 — the can task chiefly certifies that the field update preserves (and slightly extends) the base’s competence at the ceiling. The discriminative task is square, where the drifting base solves only 0.382 of episodes; Porygon adds 55 points, reaching 0.91–0.93 on every seed. The backprop actor — same base, same critic, same data, same budget — reaches 0.86–0.91 (Q1): given the identical reward signal, the field-target form extracts strictly more progress at the matched budget. Against the flow-matching pipeline, Porygon reaches parity on both tasks (0.91–0.93 vs. 0.92–0.93 on square; 0.99 vs. 0.99 on can) — but from a base less than half as strong on square (Q2): the safer update recovers the headroom the weaker base gives up.

Why does the form win? The anchor decouples two jobs that a behaviour-cloning penalty must do with one knob. Inside the dead zone the critic is followed without opposition; outside it the policy is returned. That is what lets Porygon tolerate a reward step ($\eta_Q=0.5$) far above what the backprop actor can survive, and simply takes more, safer steps in the directions where the critic is consistent. The backprop actor must instead keep its effective step small enough for its worst critic errors, and it pays in speed: Porygon reaches 0.91 on every seed by 15K steps — a level the backprop actor needs another 5–10K steps to reach. Its matched-budget endpoint trails Porygon’s on every seed, and the gap closes only near 40K, where both converge to 0.97–0.99. The form’s advantage on this benchmark is therefore a faster, never-worse path to the same asymptote — exactly what a bounded-step update should buy [TODO: quantify further with the stability-window ablation (Section 5.4, iii)].

The dial answers Q3 in its first regime: with a critic that generalizes (robomimic critics see dense, single-task data), the value-only reward fields preserve can (0.91–0.95) but collapse on square (0.53–0.58 versus Porygon’s 0.91–0.93), despite verified-healthy residuals and longer training. Deleting the gradient direction throws away exactly the signal that solves the harder task — the real-benchmark mirror of the toy’s easy regime, where the compass wins.

Table 3: LIBERO hard-8, per-task fine-tuning from the shared drifting base (success under powered evaluation: 100 rollouts $\times$ 3 evaluation seeds; Porygon rows average the last three training checkpoints, and two training seeds on t32/t53/t65). One recipe across all tasks. Across the five tasks trained with two seeds the spread is 0.000, 0.003, 0.004, 0.019 and 0.089 (t75, t73, t53, t32, t65): negligible except on the highest-headroom task, so on all tasks but t65 differences of a few points are meaningful, while on t65 only gaps above ~ 9 points are. “—” = run still in progress at draft time.

	t8	t21	t32	t53	t65	t73	t75	t81	mean
drift base (no RL)	0.818	0.690	0.610	0.785	0.573	0.648	0.578	0.588	0.661
backprop actor	0.820	0.703	0.587	0.800	0.550	0.663	0.583	0.653	0.670
Porygon ($\eta_Q=1$)	0.831	0.737	0.694	0.842	0.826	0.710	0.657	0.607	0.738
Porygon ($\eta_Q=2$)	0.857	0.752	0.729	0.853	0.812	0.752	0.648	0.635	0.755
same protocol, flow-matching base (4 tasks measured)									
FM base (no RL)	0.897	0.700	0.690	0.917	0.787	0.683	0.620	0.733	0.753
FM + DICE-RL	0.873	—	0.682	0.915	0.809	0.690	0.592	0.708	—
FM + GRPO	—	—	0.700	0.897	0.883	—	0.603	—	—
FM + Porygon	—	—	0.738	0.897	0.878	—	0.608	—	—

5.3 LIBERO hard-8: the weak-critic regime

On the eight hardest tasks of LIBERO-90 we deploy Porygon as a routing system: one frozen multitask drifting base, one small per-task residual head and critic fine-tuned by Porygon on each task’s own interactions, and task-ID routing at deployment — the architecture every published LIBERO RL fine-tuner effectively uses. One recipe is shared across all tasks with no per-task tuning ($\eta_Q=1.0$, dead-zone anchor with a dimension-independent rms radius $\rho=0.05$, 3-step returns, 40 iterations of 16 episodes).

Table 3 gives the verdict across all eight tasks: Porygon gains +3.9 to +23.9 points per task (mean +9.4 at $\eta_Q=2$, +7.7 at $\eta_Q=1$), improving on the base for every task at both settings — on the highest-headroom task it reaches 0.781, above the best previously recorded result on that task by any method — while the backpropagated actor consuming the same per-task critics averages +0.9, going below base on the two highest-headroom tasks. The robomimic ordering (Q1) transfers intact to the harder benchmark: identical reward signal, and only the field-target form converts it into gains without ever falling below the base. Two implementation findings were load-bearing and are ablated in Section 5.4: the anchor’s dead zone must be defined in dimension-independent rms units (the vector-norm form silently re-activates the anchor at LIBERO’s $S=112$ chunk dimension and freezes the residual), and the weaker critic gradient ($\|\nabla_a Q\|$ roughly $4\times$ smaller than robomimic’s) requires the doubled reward step the anchor makes safe. Sweeping the reward step across all eight tasks confirms this at the table level: $\eta_Q=2$ improves the mean from 0.738 to 0.755 and helps on six of eight tasks (the two exceptions, -0.014 and -0.009 , are inside the seed spread), and the benefit is uncorrelated with the critic’s action-gradient magnitude — the task with the deadest critic (t81, $\|\nabla_a Q\|=0.007$) gains +2.8, as much as the strongest-gradient tasks. The reward step therefore acts mainly by reducing the anchor’s drag rather than by exploiting more first-order signal, and the trust region is what keeps the larger step safe: no setting on any task falls below its base.

The flow-matching rows settle Q1 independently of the base. On its own, stronger base the state-of-the-art pipeline extracts essentially nothing per task: FM + DICE-RL scores -0.8 , -0.2 , $+2.2$ and -2.8 points relative to the FM base on t32, t53, t65 and t75 (mean -0.4), the same near-zero extraction its actor shows on the drifting base ($+0.9$). Swapping only the actor update helps on three of the four tasks: Porygon on the identical flow-matching base and budget beats FM + DICE-RL by +5.6 (t32), +6.9 (t65) and +1.6 (t75), reaching 0.738 and 0.878 on the first two — +4.8 and +9.1 over that base. The exception is instructive: on t53 the flow-matching base already scores 0.917 and neither update improves it (-1.8 for Porygon, -0.2 for DICE-RL). Indeed the never-below-base behaviour Porygon shows on all eight drifting-base tasks does not carry over to this base: on t53 and t75 it ends 2.0 and

Table 4: LIBERO hard-8 mean success under powered evaluation (100 rollouts $\times$ 3 evaluation seeds per cell; one column per training seed; 100 RL iterations). Neither base is moved by any fine-tuner: the weak multitask critic supports no extractable gain, for backpropagation and field transport alike. “—” = evaluation still running.

method	s10000	s10001	s10002
drift base (no RL)	0.671 ± 0.014		
backprop actor (same base)	0.657	0.655	0.668
Porygon (top- k)	—	0.684	—
Porygon (tilted)	0.677	—	0.672
Porygon	0.655	—	0.659
flow-matching base (no RL)	0.738 ± 0.022		
FM + DICE-RL	0.756	0.735	0.737

1.2 points below its starting point. Fine-tuning a base that is already near its ceiling is a different problem from lifting a weak one, and our trust region is calibrated for the latter.

Adding the critic-free policy-gradient baseline at the same budget completes the picture. On the one task with real headroom from the flow-matching base, GRPO and Porygon extract essentially the same amount (0.883 and 0.878 on t65, both $\approx +9.5$) while the backpropagated actor manages +2.2; on t32 Porygon leads (0.738 vs. 0.700); on the two tasks where no method gains, all three sit together. The separation on this benchmark is therefore not Porygon against the world but update form against backpropagation-through-the-critic: the field update matches what a policy-gradient method extracts, without requiring the noise-injection surrogate that makes likelihood ratios available for a one-step generator in the first place. The field-target form, not the base and not the reward signal, is what converts critic signal into success. It also puts the Q2 comparison in perspective: Porygon on the weak one-step base reaches 0.781 on t65 against the flow-matching pipeline’s 0.809 — within three points, from a base 21 points lower — and exceeds it on t75 (0.657 vs. 0.592) and on t32 once the reward step is matched to that task’s stronger critic gradient (0.729 vs. 0.682). [TODO: per-task figure; remaining FM cells and matched-budget FM + GRPO.]

Why not one shared critic? The multitask control. The alternative to routing is a single policy and critic trained on all eight tasks jointly. This setting flips the critic’s situation: our diagnostics show the shared critic’s action-gradient collapses on most tasks (per-task $\|\nabla_a Q\|$ spanning 0.003 to 0.55), and the toy predicts the resolution dial should compress.

The powered numbers (Table 4) deliver a verdict that noisy small-sample readouts had obscured: nothing wins here. On the drifting base, the backprop actor (0.655–0.668), Porygon (0.655–0.659), and every resolution variant (0.66–0.70) all sit within noise of the untuned base (0.671 ± 0.014). The flow-matching pipeline looks stronger in absolute terms (0.735–0.756) — but its own base already scores 0.738 ± 0.022 , so DICE-RL likewise extracts essentially nothing; its apparent lead on this benchmark is inherited entirely from pretraining. Applying our final recipe — every fix and hyperparameter that yields +9.4 per task — to this shared-critic setting changes nothing (0.668 versus base 0.671), confirming that what fails here is the critic, not the update rule or its tuning. Three honest conclusions follow. First, the weak-critic regime is a no-gain regime for every update rule we tested: when the shared critic’s action-signal is this poor, neither backpropagating it nor transporting along it, at any resolution, converts it into task success. Second, the dial compresses exactly as the toy predicts — gradient, ranking, and soft-value sources become indistinguishable — though in the starkest possible form. Third, every drift-base arm preserves the base within noise: the failure mode of Figure 1 does not appear even for the backprop actor, consistent with the diagnosis that this critic is action-blind rather than action-exploitable — there is no steep hallucinated gradient to ride. Small-sample evaluations told a different, wrong story (single-seed 20-episode endpoints ranged 0.63–0.78 across these same arms); powered evaluation collapses that spread to the base value. [TODO: per-task breakdown figure (scripts/plot_libero_pertask.py) once the remaining seed evaluations land; include the guarded BC-filter interaction in the appendix.]

[Figure reserved: two ablation panels.] Left — stability window: final square success vs. reward step η_Q for Porygon and for the backpropagated control (A + gradient clipping); the claim is a wide flat optimum for the field form vs. a narrow peak for backpropagation. Right — dispersion retention: action dispersion and best-of- N curve before/after fine-tuning for Porygon vs. A.

Figure 5: Ablations: where the trust region and update form pay. [TODO: runs in progress]

Table 5: LIBERO ablations: leave-one-out on the frozen recipe, two tasks, powered evaluation (last-3 checkpoints). Every row changes exactly one component. Δ is relative to the full recipe.

variant	t65 (base 0.573)		t32 (base 0.610)	
	success	Δ	success	Δ
full recipe	0.781	—	0.684	—
– dimension-independent anchor (vector-norm ρ)	0.662	−0.119	0.655	−0.029
– dead zone ($\rho=0$)	0.549	−0.232	0.601	−0.083
– doubled reward step ($\eta_Q=0.5$)	0.769	−0.012	0.615	−0.069
– 3-step returns ($n=1$)	0.782	+0.001	0.512	−0.172
wider trust region ($\delta_{\text{tot}}=0.25$)	0.874	+0.093	0.689	+0.005
stronger reward step ($\eta_Q=2.0$)	0.802	+0.021	0.729	+0.045
reward field $\rightarrow$ top- k (no gradient)	0.577	−0.204	0.341	−0.343
reward field $\rightarrow$ tilted (no gradient)	0.697	−0.084	0.378	−0.306

5.4 Ablations: which component pays where

Table 5 isolates each component. Three findings matter. First, the anchor geometry is load-bearing: removing the dead zone entirely ($\rho=0$) drops t65 to 0.549, below the un-tuned base — the un-gated restoring pull erases the residual, the quantitative version of the toy’s “penalties lose to fields” lesson — and measuring the dead-zone radius in vector-norm rather than dimension-independent units costs 11.9 points, because at LIBERO’s $S=112$ chunk dimension the base residual already exceeds ρ at initialisation, so the anchor is active from the first update.

Second, and most decisively, deleting the critic’s gradient direction is fatal, and the effect replicates. On t65 top- k transport falls to 0.577 and tilted to 0.697; on t32 the same two substitutions collapse to 0.341 and 0.378 — not merely below the full recipe but far below the un-tuned base (0.610), a 34- and 31-point deficit. Two tasks, two bases (Table 1 shows the same ordering on robomimic square), one conclusion: the ranking information a critic can supply is not a substitute for its gradient, and Q3 is answered the same way wherever we ask it. This is what separates the method from value-only transport such as critic-ranked top- K drift.

Third, the remaining knobs are task-dependent, and in a way that tracks the critic rather than the task’s difficulty. On t32, whose critic carries a strong action-gradient ($\|\nabla_a Q\| \approx 0.55$), 3-step returns are worth 17 points and a doubled reward step 4.5 (0.729), while widening the movement bound alone ($\delta_{\text{tot}}=0.25$) does nothing (+0.5): this task wants more reward signal, not more room to move. On t65, whose gradient is an order of magnitude weaker (≈ 0.035), the pattern inverts — 3-step returns and the doubled step are within noise, but widening the trust region is worth +9.3 points (0.874), the single largest positive ablation we measure. We report this as a regime difference rather than a tuned optimum: it is one seed, and t65 carries a ± 9 -point seed spread (Section 6), so the direction is trustworthy and the magnitude is not. What a bounded-step field update needs is therefore not one universal

Table 6: Dead-zone radius sweep, single seed, powered evaluation (last-3 checkpoints). $\rho=0$ is an always-on behaviour-cloning penalty; $\rho \rightarrow \infty$ removes the anchor entirely. Performance is flat across two orders of magnitude of ρ and collapses only at the endpoints, so what matters is that a dead zone exists, not its precise size.

dead-zone radius ρ	t65 (base 0.573)	t32 (base 0.610)
0 (always-on penalty)	0.549	0.601
0.02	0.863	0.734
0.05 (recipe)	0.781	0.684
0.10	0.812	0.694
0.20	0.771	0.671
∞ (no anchor)	0.803	0.671
no anchor and no step bound	0.000	0.000

setting but one matched to what the critic actually carries — and the trust region is what makes raising either knob safe to attempt.

Does the size of the dead zone matter? Table 6 sweeps ρ over two orders of magnitude. Every interior setting lands between 0.771 and 0.863 on t65 and between 0.671 and 0.734 on t32 — a spread no larger than this benchmark’s seed-to-seed variation — while the $\rho=0$ endpoint, which is exactly an always-on behaviour-cloning penalty, falls to roughly base level on both tasks. Removing the anchor entirely ($\rho \rightarrow \infty$) is not similarly fatal, because the step bound is still present. Removing both is: performance is exactly 0.000 on both tasks. We read this as follows. Some constraint tying the policy to its base is necessary, and its absence is catastrophic rather than merely costly. Beyond that, at this budget we cannot distinguish between the available constraints, and we do not claim a preferred radius; $\rho=0.05$ is a choice inside a wide flat region, not an optimum.

How often does the bound actually bind? It would be easy to over-read our own mechanism, so we measured it. The clip in Equation (4) only ever reduces a displacement, so whenever a particle’s proposed motion is already shorter than δ_{tot} the update is exactly the unbounded one, and the critic is still setting the distance for that particle. In the Section 3.3 setting, at our operating point ($\delta_{\text{tot}}=0.15$, $\eta_Q=1$), the clip is active on 17.3% of sampled actions, and the median proposed displacement is only $0.12\delta_{\text{tot}}$. Tightening the cap to 0.05 raises the binding rate to 26.3%; loosening it to 1.0 drops it to 4.8%.

These numbers are worth stating because they rule out the reading we initially favoured. If the cap were the mechanism, it would have to be doing its work through the 17% of proposals it truncates — yet removing the cap entirely, while keeping the restoring force, leaves the method unchanged, and keeping the cap while removing the restoring force destroys it (Figure 1b). The cap is a numerical safeguard. What holds the policy on the data is the anchor.

Is the gap just a better step size? The most economical explanation of Table 3 would be that we simply tuned our update’s step better than the baseline’s. We tested that directly on t65 by sweeping each method’s own step-size knob under the identical evaluation protocol, and the explanation fails. Sweeping the backpropagated actor’s learning rate over a $10\times$ range gives 0.550 (10^{-4}), 0.548 (3×10^{-4}) and 0.578 (10^{-3}) — every setting at or below the un-tuned base (0.573). Sweeping our reward step over an $8\times$ range gives 0.775, 0.817, 0.908 and 0.750 for $\eta_Q \in \{1, 2, 4, 8\}$ — every setting 18–33 points above that same base. The two bands do not overlap at any step size either method was given. A larger step does not convert the backpropagated update into the field update; the difference is the form, and it is what makes a large step survivable at all. (Both sweeps are single-seed and the within-band ordering should not be read as a tuned optimum — the $\eta_Q=4$ peak sits close to this task’s seed spread. The band separation, at 18 points minimum, does not.)

[TODO: Remaining ablation plots (Figure 5); robomimic-side items below: (i) trust-region locus — the backprop actor + parameter-space gradient clipping vs. B’s action-space clip

at equal effective step (isolates the central claim); (ii) anchor geometry — $\rho=0$ silently erases the residual under weak reward fields (measured residual RMS collapses $\sim 5\times$ and performance reverts to base-plus-selection), the quantitative version of the toy’s “penalties lose to fields” lesson; (iii) step/clip sweep — the 21-configuration square sweep behind the family recipe: stability window of η_Q under Equation (4) vs. backpropagation; (iv) dispersion retention — action dispersion and best-of- N gains after fine-tuning, vs. control (does the field update preserve the multimodality the base was trained for?).]

6 Conclusion

We showed that the natural update primitives for RL fine-tuning of one-step generative policies are, respectively, unavailable without modifying the policy and unsafe when the critic is imperfect, and that the policy class’s own pretraining primitive — field-target regression — provides the missing update rule. Porygon expresses reward and anchoring as composable velocity fields over action particles, and recovers the backpropagated direction in the small-step limit. Its working component is the action-space trust region: the critic sets the direction, while a bound — a pointwise clip on each particle’s displacement, a restoring pull outside a radius of the frozen base, or both — sets how far any single update may carry the policy.

The experiments locate the advantage precisely. Given the same base, critic, data and budget, the field-target form extracts far more than backpropagating that critic: +55 points on robomimic square, and +9.4 versus +0.9 on average across the eight hardest LIBERO-90 tasks, where the backpropagated actor falls below its base on the two highest-headroom tasks and Porygon improves on every one. The same ordering holds when the base is swapped for the stronger flow-matching policy, so the effect is attributable to the update rule rather than to the starting point. Against a critic-free policy-gradient baseline at matched interaction budget, Porygon is at parity (+10.1 versus +9.5 per-task mean on LIBERO; 0.878 versus 0.883 on the flow-matching base’s one high-headroom task) — while requiring none of the noise injection that makes likelihood ratios definable for a deterministic one-step generator in the first place. Ablations confirm the mechanism rather than the hyperparameters, and they constrain what we may claim. Replacing the dead-zone with an always-on penalty ($\rho=0$, exactly a behaviour-cloning term) falls back to base level (0.549 and 0.601 against bases of 0.573 and 0.610), so the constraint must be shaped in action space rather than added to the loss. Switching the two guards on and off independently is decisive in one direction only: removing both collapses success to exactly 0.000 on both tasks, while either alone recovers essentially all of the gain — indeed removing the anchor and keeping only the step bound is among our strongest settings (0.803 on t65). We therefore claim the necessity of an action-space constraint, and not the superiority of the particular one we ship.

Limitations. Three specific results run against our thesis and we state them plainly. On t81 the backpropagated actor ends above Porygon (0.653 versus 0.635); on two flow-matching tasks the un-finetuned base beats every fine-tuner including ours (0.917 on t53, 0.620 on t75); and on robomimic can the flow-matching pipeline edges us on two of three seeds. Single-seed variation is also substantial on LIBERO — repeating the frozen recipe on t65 with a second seed moves the result by 8.9 points — so per-task deltas smaller than roughly ten points, including our η_Q and δ_{tot} sweeps, indicate direction rather than magnitude. The claims that survive this variation are the ones separated by margins far larger than it: the +9.4 versus +0.9 mean gap, the non-overlapping step-size bands, and the below-base collapse when the critic’s gradient is deleted.

The gains track how much a critic actually knows. When a single critic is shared across all eight LIBERO tasks its action-gradient collapses and no critic-based update — ours or the state of the art’s — improves on its base; when a base is already near its ceiling, Porygon can end slightly below it (−2.0 and −1.2 on two flow-matching tasks), since our trust region is calibrated for lifting weak bases rather than for refining strong ones. Our critic-trust diagnosis is also post hoc: choosing the reward-field resolution online from critic-trust signals is the natural next step, as is testing whether the reward step should adapt per task. Finally, all results are state-based simulation on a single embodiment.

We should be precise about which part of the story generalizes, because two claims are easy to conflate. The update rule is base-agnostic: it consumes a frozen noise-to-action map, a residual head and a critic, and never touches how the base draws its action — which is why it transfers unchanged to a flow-matching base, where it also improves on the state-of-the-art fine-tuner. What does not generalize is the economy of the construction. A drifting model is pretrained by field-target regression, so on that base the fine-tuner reuses machinery the policy class already has; a flow-matching model is pretrained by conditional velocity regression, so there the same update is bolted on rather than inherited (and costs one base forward pass per particle per integration step). The principle we can defend is therefore narrow: what keeps a critic-driven update on the data is a restoring force with a dead zone, not a limit on step size. The distinction is not cosmetic. A step limit constrains speed, and a drift assembled from permitted small steps still arrives; in our isolation the limit alone leaves the policy at zero success. A behaviour-cloning penalty restores, but does so everywhere, including the region where the critic is reliable, and must therefore trade far-field containment against near-field progress with a single coefficient; it also fails alone. The dead zone separates those jobs.

We deliberately claim no more than this. We have no principled rule for choosing ρ — it is tuned, not derived — and the argument for why the dead zone should exist is an argument about matching the regularizer’s spatial profile to the critic’s reliability, not a bound. Deriving ρ from a measure of local critic reliability is the obvious next question, and we leave it open.

Reproducibility statement

[TODO: Code release (training framework + official-harness implant), exact configs (family-shared recipe), seeds, eval protocols (300-ep robomimic; 100×3 powered LIBERO), compute (single L40S/A100 per run, 8–20h), toy experiment script. All numbers in Table 1 are averages, never best-of.]

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A Additional results

[TODO: Full seed tables, training curves, hard-8 per-task breakdown.]

B Hyperparameters

[TODO: Family-shared recipe table; official DICE-RL hypers; sweep grids.]

C The kernel transport operator

[TODO: Full form of $\mathcal{V}$: global distance scale, per-radius softmax-softmax affinity, contested-attractor weighting, self-exclusion, per-radius normalization on/off (restoring vs. constant-pull), radii $\{0.02, 0.05, 0.2\}$. Port from `drift_field.py` docstrings.]

D Toy two-regime construction

[TODO: Details of the hard-critic/easy-critic toy; failure of compass updates.]